MAKERERE UNIVERSITY

SCHOOL OF COMPUTING AND INFORMATICS TECHNOLOGY

UNIVERSITY EXANINATIONS 2022/2023

SEMESTER II

PROGRAM:	BSC COMPUTER SCIENCE
COURSE CODE:	CSC 1201
COURSE NAME:	PROBABILITY AND STATISTICS
DATE:	14th JUNE 2023
DURATION:	3 HRS
START:	12:00 NOON
END:	3:00 PM

INSTRUCTIONS:

1. Attempt all questions in Section A
2. Attempt any three questions in Section B
3. Start each Section B question on a fresh sheet
4. Write neatly

SECTION A

(a) State any two measures of dispersion of data (2 marks)

(b) With a brief reason, state whether or not the following statements are correct with respect to random variables (2 marks each)

i. A mean of a descrete random variable is always one of the possible outcomes
ii. For a continuous random variable, the probability of an outcome being a single value is 0
iii. Every random variable where the mean, the mode and the median are the same follows a normal distribution
iv. A uniform distribution may be descrete or continious

(c) A "rudo" player toses a fair dice and starts to play when a 6 comes on top. He has 4 chances to toss per round and on failing, he wasits for another round. Find the probability that

i. He gets starts to play on the final attempt (3 marks)
ii. He does not start to play after a round (2 marks)

(d) With a brief reason, state which of the following statements are true for any two events A and B

i. If A and B are independent, then A' and B' are independent (3 marks)
ii. $P(A/B) < P(A)$ (3 marks)
iii. $P(A \cap B) < P(AUB)$ (3 marks)
iv. $P(A/B) = P(B/A)$ (3 marks)

(e) It is known that 30% of the population has covid. The covid testing reaget has some innacuracies. It will declare 2% of infected persons uninfected while 3% of the people declared uninfected are actually covid +. what is the probability that

i. the deduction made from the test is actually true (4 marks)
ii. a person who is declared negative is actually positive (4 marks)

(f) Two students Sseeki and Sseeka agreed to meet at a place to revise but forgot. Sseki toses a coin to decide whether to go to library or not and in the later case he toses a coin to decide whether he goes to Big Lectur room or small lecture room. Sseeka toses a coin to decide to go to Library or large lecture room. Find the probability that

i. they meet (2 marks)
ii. they go to different places (3 marks)

SECTION B

Question One

The data for results of statistics test for 40 students was collected as below.

16, 17, 18, 3, 1, 23, 18, 13, 10, 21, 1, 1, 13, 21, 13, 15, 19, 24, 16, 2, 23, 5, 12, 18, 8, 12, 6, 8, 16, 3, 3, 5, 0, 7, 9, 12, 20, 10, 2, 23

0 - 4 = 6.
5 - 9 = 10
10 - 14 = 8
15 - 19 = 9.
20 - 24 = 7

Use the data to answer the questions below.

(a) Find the range of the data (2 marks)

(b) Using a class width of 5 (starting with 0-4 class), generate a frequency distribution table for the data (4 marks)

(c) Use the grouped data to compute the mean, variance, mode and Inter quartile range for the data (14 marks)

Question Two

(a) A continuous random variable is described as $f(x) = \lambda e^{-x}$ and defined over $0 < x < \infty$

i. Show that $\lambda = 1$ (3 marks)
ii. Find $P(1 < x < 10)$ (3 marks)
iii. Generate and sketch the cummulative distributionfunction $F(x)$ (6 marks)

(b) The lifespan of IBM external drives is known to be normally distributed with mean 10 years and standard deviation δ. IBM is willing to replace the worst performing 3% of the drives. The company set a replacement guarantee of 9.2 years. Compute the value of δ (8 marks)

Question Three

(a) When a CPU overheats, a Desktop computer will switch off or crash. While switch off is recoverable, crush is not. The overheat of the CPU may or may not be caused by fan failure. Chances of a fan failure are 1.5%. Once a fan has failed, the computer will have to either switch off (65% of cases) or crash (35% of cases). However, with a well functioning fan, CPU can still over heat with 1% chances of crush.

i. Represent this information on a probability tree diagram (5 marks)

ii. Find the probability of a CPU crashing (5 marks)

iii. If data from the company show that there were 1345 CPUs that crashed, how many are in circulation (4 marks)

(b) With brife precise illustrations, describe the meaning of

i. Independent events (2 marks)

ii. Mutually exclusive events (2 marks)

iii. Conditional even occurance (2 marks)

Question Four

(a) A random variable X is generated by tossing two fair dice D_1 and D_2 together and the outcome is the absolute difference of the average of the two faces is from 3. ie x= | average(D_1, D_2) -3|.

i. Generate the probability densty function for X (7 marks)

ii. Find the mean, mode and median for X (6 marks)

(b) The probability that a person has Hepatitis B is 0.2. A person is picked from the population and tested for Hepatitis B. When a positive test is got, testing stops

i. Find the expected number of tests (3 marks)

ii. Find the probability thatmore than 3 tests will be made (4 marks)